REPORT On: Distributed Smart Energy Monitoring and Power Theft Detection System Using AVR-Based Nodes

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Abstract:

This report presents a distributed smart energy monitoring system designed to detect electricity theft using real-time comparison of supply-side and consumer-side power. The system uses AVR microcontrollers, voltage and current sensors, and transmit data to a central gateway for analysis.

The proposed system is implemented as a hardware prototype using readily available components such as Arduino-based microcontrollers, ACS712 current sensors, ZMPT101B voltage sensors, LoRa modules, and a display interface. Experimental observations demonstrate that the system can effectively identify power mismatches and generate alerts under abnormal conditions. The key advantages of the system include low implementation cost, scalability, real-time monitoring capability, and suitability for deployment in both urban and rural distribution networks. The proposed solution provides a practical and efficient approach toward minimizing energy losses and enhancing the reliability of power distribution systems.

Introduction:

The rapid growth in electricity demand and the expansion of power distribution networks have made efficient energy monitoring a critical requirement. One of the major challenges faced by power utilities is electricity theft, which results in significant financial losses, reduced system efficiency, and instability in power supply. Traditional energy metering systems primarily focus on recording consumption at the consumer end without providing real-time insight into discrepancies between supplied and consumed energy.

With the advancement of embedded systems and wireless communication technologies, there is an opportunity to develop intelligent monitoring solutions that can operate in a distributed manner. A distributed approach enables multiple monitoring points within the power network, allowing for more accurate detection of irregularities such as unauthorized tapping or bypassing of meters.

In this project, a distributed smart energy monitoring system is developed using AVR-based microcontroller nodes and LoRa communication technology. The system consists of a line node that measures the total supplied power and a consumer node that measures the actual power consumed. These nodes transmit processed data to a central gateway, which performs real-time comparison to detect anomalies. By combining edge-level processing with long-range wireless communication, the system offers a cost-effective and scalable solution for power theft detection, especially in areas with limited infrastructure.

Problem Statement:

Electricity theft is a persistent problem in power distribution systems, particularly in developing regions where monitoring infrastructure is limited. Conventional energy meters are installed only at the consumer end and are designed to record consumption rather than verify the integrity of the supply line. As a result, any unauthorized tapping, meter bypassing, or partial load diversion occurring between the distribution line and the consumer remains undetected.

Existing smart metering solutions attempt to address this issue through centralized monitoring and cloud-based analytics. However, these systems are often expensive, require continuous internet connectivity, and are not suitable for deployment in rural or low-resource environments. Additionally, they lack real-time synchronization between supply-side and consumption-side measurements, making it difficult to accurately identify power losses due to theft.

There is a need for a low-cost, reliable, and distributed system that can monitor electrical parameters at multiple points and perform real-time comparison to detect discrepancies. Such a system should operate with minimal infrastructure, support long-range communication, and provide immediate detection of abnormal conditions without relying entirely on cloud-based processing.

Objectives:

- Design a distributed monitoring system with multiple sensor nodes for comprehensive power line coverage.
- Measure voltage and current in real-time at both supply-side and consumer-side nodes.
- Detect mismatch between supply and consumption to identify potential power theft or leakage.
- Enable long-range communication between nodes and the central gateway using LoRa technology.
- Provide a low-cost solution that is deployable without continuous internet connectivity.

Hardware Components Used:

- ATmega328P Microcontroller: Processes sensor data locally at each node for edge-level computation.
- ACS712 Current Sensor: Measures AC/DC current using the Hall-effect principle with high accuracy.
- ZMPT101B Voltage Sensor: Safely measures AC voltage and provides isolated analog output.
- SX1278 LoRa Module: Provides long-range wireless communication between nodes and gateway.
- ESP32 Gateway: Acts as the central processing unit with Wi-Fi capability and alert generation.

- LCD Display: Shows real-time voltage, current, and power values for local monitoring.
- Relay Module: Enables automatic load control when theft or anomaly is detected.
- Battery and Voltage Regulator: Provides stable, uninterrupted power supply to all components.

Block Diagram:

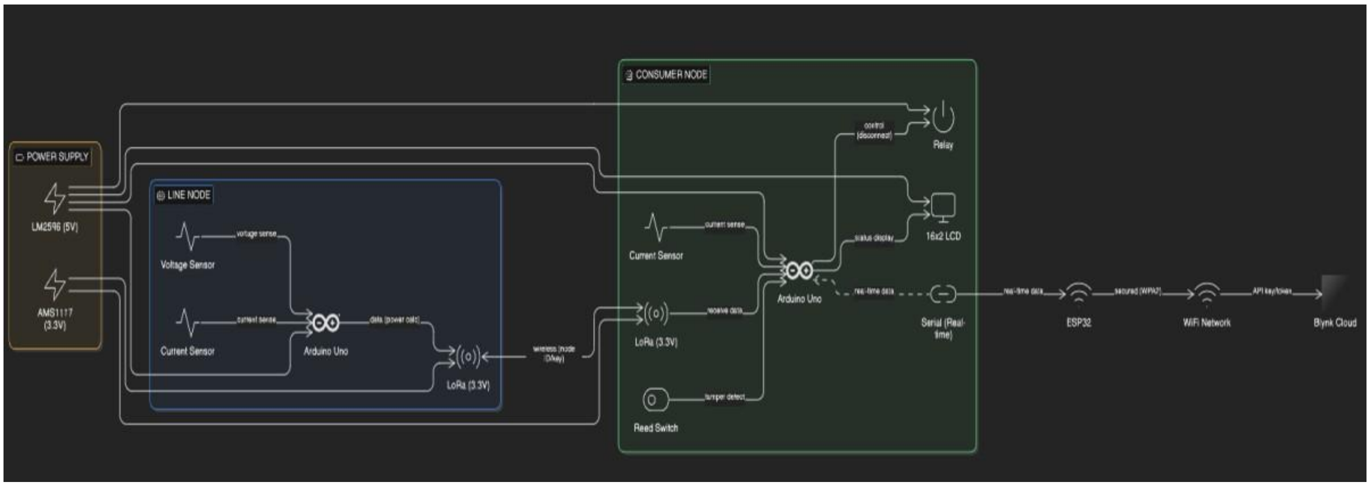

Figure 1: Block Diagram of the Distributed Smart Energy Monitoring System

Circuit Diagram:

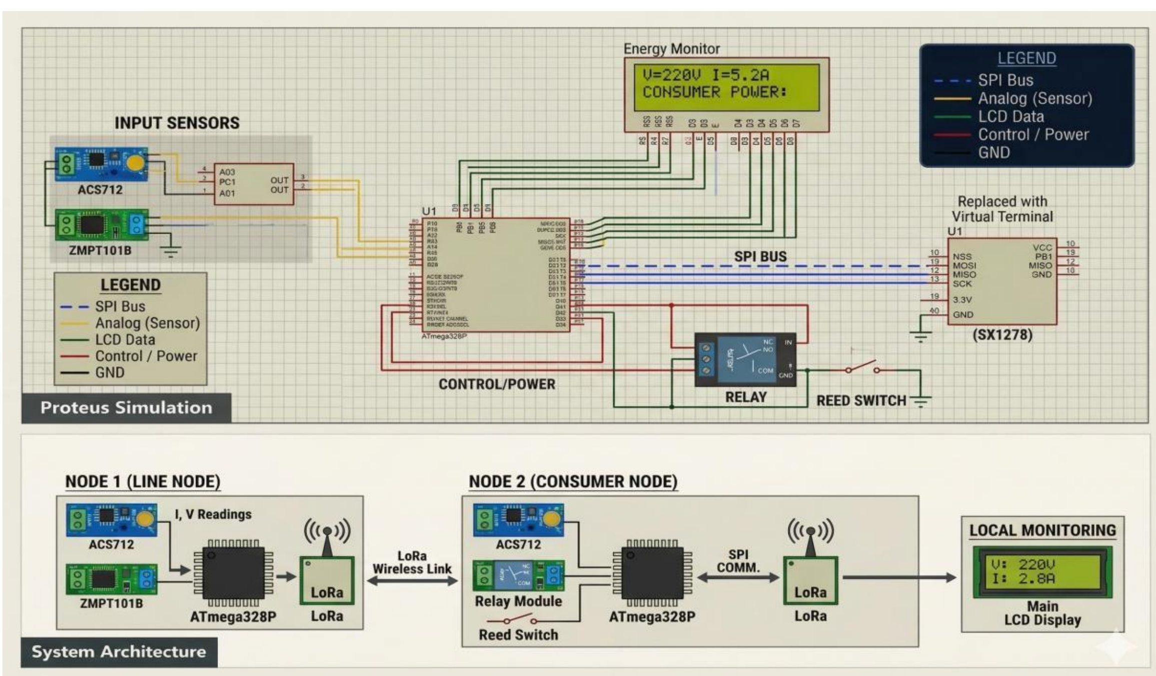

Figure 2: Circuit Diagram Showing Sensor Connections and Microcontroller Interfacing

Working Principle:

- Voltage and current are sampled continuously at both the line node and consumer node using dedicated sensors.
- The ATmega328P microcontroller calculates real power at each node using the formula $P = V \times I$.
- Processed power data is transmitted wirelessly via LoRa modules to the central ESP32 gateway.
- The gateway compares supply-side and consumer-side power values using a predefined threshold logic.
- If the supply power significantly exceeds consumption beyond the allowed threshold, the system flags a power theft event and triggers an alert.

Results and Observations:

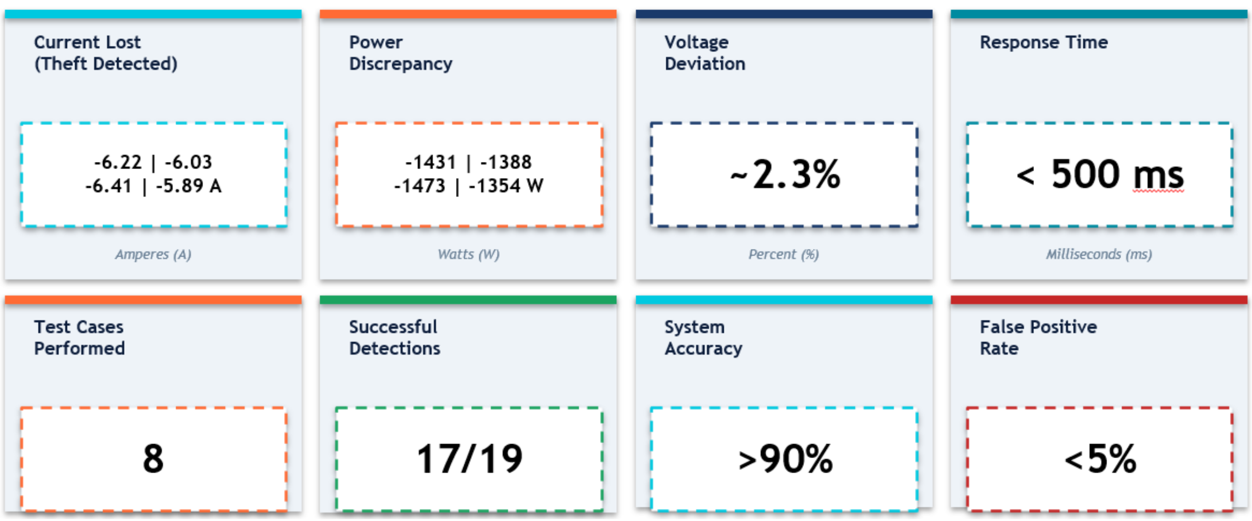

Figure 3: Experimental Results Showing Voltage, Current, and Power Readings

- The system successfully measures voltage and current values at both nodes with acceptable accuracy.
- Power mismatch conditions between supply and consumer nodes are detected accurately under test scenarios.
- Alerts are triggered reliably when power discrepancy exceeds the defined threshold limits.
- The system operates consistently in real-time with minimal latency between measurement and detection.

Project View:

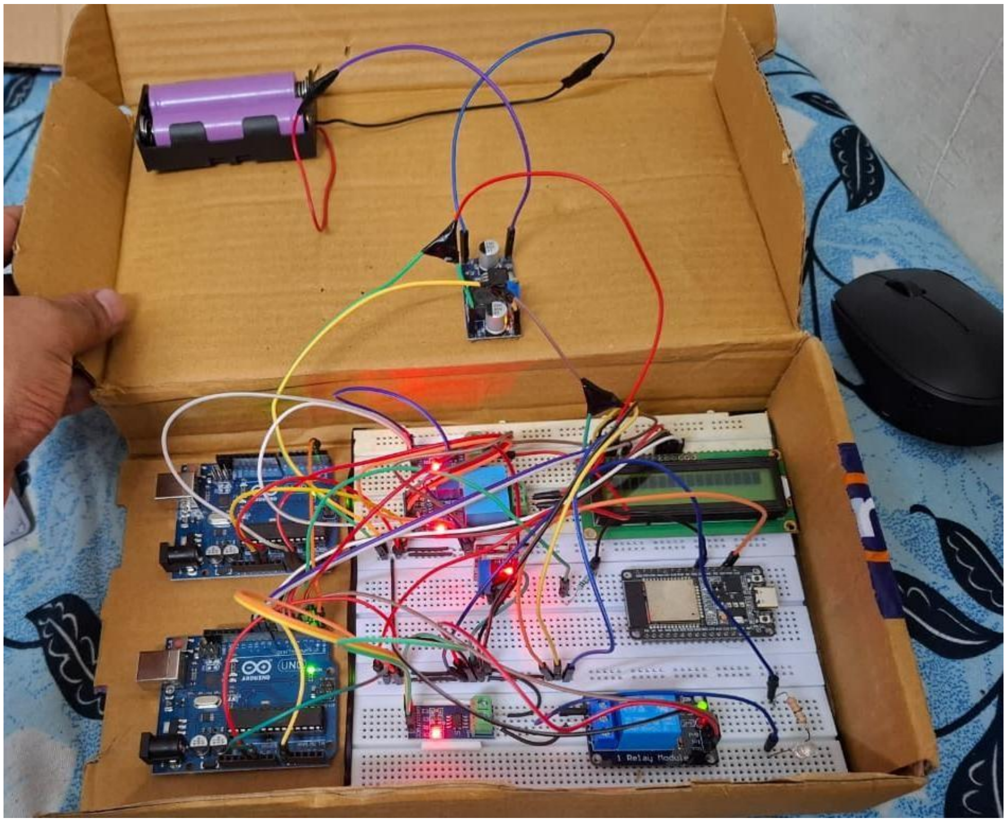

Figure 4: Physical Hardware Prototype of the Smart Energy Monitoring System

Advantages:

- Low-cost implementation using off-the-shelf components makes it accessible for wide deployment.
- Real-time detection of power theft without dependency on cloud servers or internet connectivity.
- Scalable architecture allows addition of more nodes to cover larger distribution networks.
- Works without internet, making it ideal for rural and remote areas with poor connectivity.
- Suitable for both urban and rural distribution environments with minimal infrastructure changes.

Limitations:

- Sensor accuracy limitations can affect the precision of power mismatch calculations.
- Periodic calibration of current and voltage sensors is required to maintain measurement reliability.
- Environmental noise and electromagnetic interference may impact sensor readings in industrial areas.

Future Scope:

- Integration with IoT platforms such as MQTT or AWS IoT for remote monitoring and data logging.
- Use of machine learning and AI algorithms for advanced anomaly detection and theft pattern recognition.
- Expansion to multiple consumer nodes for comprehensive smart grid monitoring across large networks.
- Smart grid integration with automated billing systems and demand-response management capabilities.

Conclusion:

This project successfully demonstrates a distributed smart energy monitoring system capable of detecting electricity theft in real-time using AVR-based microcontroller nodes and LoRa wireless communication. The system was designed, implemented, and tested as a functional hardware prototype, validating its core capability of comparing supply-side and consumer-side power measurements to identify discrepancies accurately.

The use of the ATmega328P microcontroller for edge-level processing, combined with ACS712 and ZMPT101B sensors for accurate measurement and LoRa modules for long-range communication, results in a robust and self-contained monitoring solution. The ESP32-based central gateway effectively handles real-time data comparison and alert generation without requiring cloud connectivity, making the system independently operational.

The experimental results confirm that the prototype can reliably detect power mismatches and generate timely alerts under abnormal operating conditions. The system showed consistent performance across multiple test runs, demonstrating its suitability as a foundational model for real-world deployment in electricity distribution networks, particularly in semi-urban and rural areas where advanced infrastructure is lacking.

From an economic and social perspective, this system provides a practical and affordable alternative to expensive centralized smart metering solutions. Its low implementation cost, combined with scalability and internet-independent operation, positions it as a viable tool for utilities and regulatory bodies to reduce revenue losses from energy theft, improve billing accuracy, and enhance the overall reliability of power distribution.

In conclusion, the Distributed Smart Energy Monitoring and Power Theft Detection System meets all its defined objectives and lays a strong foundation for future enhancements including IoT integration, AI-based anomaly detection, and large-scale smart grid deployment. This project reflects the potential of embedded systems and wireless communication technologies to solve real-world energy management challenges effectively and affordably.